

PRINCIPLES OF GEOGRAPHIC INFORMATION SYSTEM SEM : VI

SEM VI : UNIT 5



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1. What is the relationship between Map and GIS?**(APR-19)****Ans:**

There is a strong relationship between maps and GIS. More specifically, maps can be used as input for a GIS.

-Where

-What

-When

As soon as a question contains a “where?” question, a map can often be the most suitable tool to solve the question and provide the answer. “Where do I find Enschede?” and “Where did ITC’s students come from?” are both examples. Of course, the answers could be in non-map form like “in the Netherlands” or “from all over the world.” These answers could be satisfying, however, they do not give the full picture.

A map would put these answers in a spatial context. It could show where in the Netherlands Enschede is to be found and where it is located with respect to Schiphol–Amsterdam airport, where most students arrive. A world map would refine the answer “from all over the world,” since it reveals that most students arrive from Africa and Asia, and only a few come from the Americas, Australia and Europe.

As soon as the location of geographic objects (“where?”) is involved, a map becomes useful.

However, maps can do more than just providing information on location. They can also inform about the thematic attributes of the geographic objects located in the map. An example would be “What is the predominant land use in southeast Twente?” The answer could, again, just be verbal and state “Urban.” However, such an answer does not reveal patterns. Maps can answer the “What?” question only in relation to location.

A third type of question that can be answered from maps is related to “When?” For instance, “When did the Netherlands have its longest coastline?” The answer might be “1600,” and this will probably be satisfactory to most people. However, it might be interesting to see how this changed over the years. A set of maps could provide the answer.



2. Explain the visualization process in GIS.

(APR-19)

Ans:

The cartographic visualization process is used for translation or conversion of spatial data from a database into graphics.

The producer of these visual products may be a professional cartographer, but may also be a discipline expert, for instance, mapping vegetation stands using remote sensing images, or health statistics in the slums of a city. To enable the translation from spatial data into graphics, we assume that the data are available and that the spatial database is well-structured.

Visualizations can be, and are, created during any phase of the spatial data handling process as indicated before. They can be simple or complex, while the production time can be short or long.

The visualization process is guided by the question “How do I say what to whom?” “How” refers to cartographic methods and techniques. “I” represents the cartographer or map maker, “say” deals with communicating in graphics the semantics of the spatial data. “What” refers to the spatial data and its characteristics, (for instance, whether they are of a qualitative or quantitative nature). “Whom” refers to the map audience and the purpose of the map—a map for scientists requires a different approach than a map on the same topic aimed at children.

The characteristic of maps and their function in relation to the spatial data handling process was explained in the previous section. In this context the cartographic visualization process is considered to be the translation or conversion of spatial data from a database into graphics. These are predominantly map like products. During the visualization process, cartographic methods and techniques are applied. These can be considered to form a kind of grammar that allows for the optimal design and production for the use of maps, depending on the application.

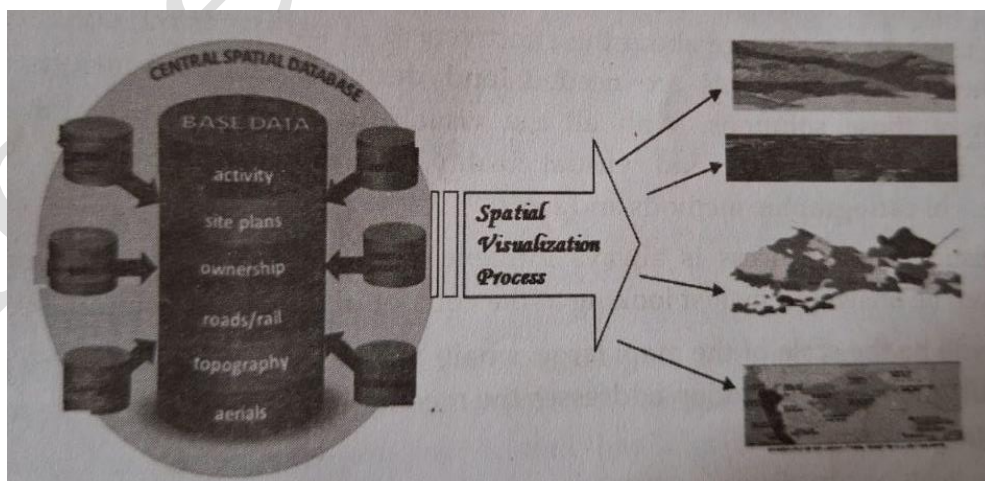


Figure: Spatial Visualization

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The visualization process can vary greatly depending where in on the spatial data handling process it takes place and the purpose for which it is needed. Visualizations can be, created during any phase of the spatial data handling process. They can be simple or complex, while the production time can be short or long.

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3. How to map terrain elevation? Explain.

(APR-19)

Ans:

Terrain elevation can be mapped using different methods. Often, one will have collected an elevation data set for individual points like peaks, or other characteristic points in the terrain. Obviously, one can map the individual points and add the height information as text. However, a contour map, in which the lines connect points of equal elevation, is generally used. To visually improve the information content of such a map the space between the contour lines can be filled with color and value information following a convention, e.g. green for low elevation and brown for high elevation areas. This technique is known as hypsometric or layer tinting. Even more advanced is the addition of shaded relief. This will improve the impression of the three-dimensional relief.

The shaded relief map uses the full three-dimensional information to create shading effects. This map, represented on a two-dimensional surface, can also be floated in three-dimensional space to give it a real three-dimensional appearance of a 'virtual world', as shown in Figure (d). Looking at such a representation one can immediately imagine that it will not always be effective. Certain (low) objects in the map will easily disappear behind other (higher) objects.

Interactive functions are required to manipulate the map in three-dimensional space in order to look behind some objects. These manipulations include panning, zooming, rotating and scaling. Scaling is needed, particularly along the z-axis, since some maps require small-scale elevation resolution, while others require large-scale resolution, ie. vertical exaggeration. One can even imagine that other geographic, three-dimensional objects (for instance, the built-up area of a city and individual houses) have been placed on top of the terrain model, like it is done in Google Earth. Of course, one can also visualize objects below the surface in a similar way, but this is more difficult because the data to describe underground objects are sparsely available.

Socio-economic data can also be viewed in three dimensions. This may result in dramatic images, which will be long remembered by the map user. Figure (E) shows the absolute population figures in three dimensions. Instead of proportionally sized circles to depict the number of people living in a region the proportional height of a region now indicates total population. The image clearly shows the large column on the right is the highest populated slum regions in Mumbai.

4. What are Bertin's six categories of visual variables?

(APR-19)(NOV-19)(APR-23)

Ans:

Basic elements of a map, irrespective of the medium on which it is displayed, are point symbols, line symbols, area symbols, and text. The appearance of point, line, and area symbols can vary depending on their nature. Most maps in this book show symbols in different size, shape and colour. Points can vary in form or colour to represent the location of shops or they can vary in size to represent Symbolology aggregated values (like number of inhabitants) for an administrative area. Lines can vary in colour to distinguish between administrative boundaries and rivers or vary in shape to show the difference between railroads and roads.

Areas follow the same principles: difference in colour distinguishes between different vegetation stands.

The variations in symbol appearance are only limited by the imagination they can be grouped together in a few categories. Bertin distinguished six categories, which he called the visual variables, and which may be applied to point, line and area symbols.

differences in:	symbols		
	point	line	area
size			
value			
grain			
colour			
orientation			
shape			

5. How to distinguish between three temporal cartographic techniques?

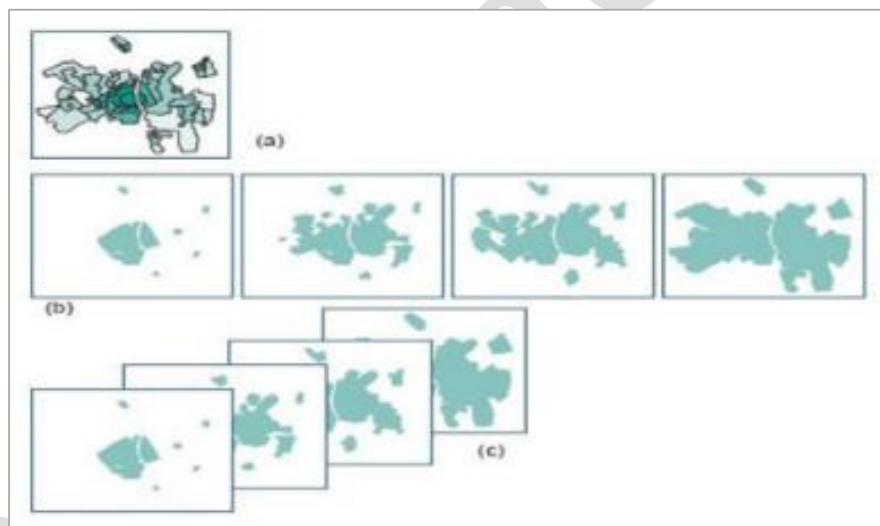
Explain.

(APR-19)(NOV-19)

Ans:

Temporal cartographic techniques:

1. **Single static map:** Specific graphic variables and symbols are used to indicate change or represent an event. Figure(a) applies the visual variable value to represent the age of the built-up areas;
2. **Series of static maps:** A single map in the series represents a 'snapshot' in time. Together, the maps depict a process of change. Change is perceived by the succession of individual maps depicting the situation in successive snapshots. It could be said that the temporal sequence is represented by a spatial sequence, which the user must follow, to perceive the temporal variation. The number of images should be limited since it is difficult for the human eye to follow long series of maps (Figure(b));
3. **Animated map:** Change is perceived to happen in a single image by displaying several snapshots after each other just like a video cut with successive frames. The difference with the series of maps is that the variation can be deduced from real 'change' in the image itself, not from a spatial sequence (Figure (c))



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6. Write a note on Map Cosmetics.

(APR-19)

Ans:

Most maps are correct from a cartographic grammar perspective. However, many of them lack the additional information needed to be fully understood that is usually placed in the margin of printed maps. Each map should have, next to the map image, a title, informing the user about the topic visualized. A legend is necessary to understand how the topic is depicted. Additional marginal information to be found on a map is a scale indicator, a north arrow for orientation, the map datum and map projection used, and some lineage information, (such as data sources, dates of data collection, methods used, etc.). Further information can be added that indicates when the map was issued, and by whom (author / publisher). All this information allows the user to obtain an impression of the quality of the map and is comparable with metadata describing the contents of a database or data layer.

- **Space constraints**

On paper maps, these elements (if all relevant) must appear next to the map face itself. Maps presented on screen often go without marginal information, partly because of space constraints. However, on-screen maps are often interactive, and clicking on a map element may reveal additional information from the database. Legends and titles are often available on demand as well.

- **Text**

Text is used to transfer information in addition to the symbols used. This can be done by the application of the visual variables to the text as well. Another common example is the use of colour to differentiate (at nominal level) between hydrographic names (in blue) and other names (in black). The text should also be placed in a proper position with respect to the object to which it refers.

- **Contrast and visual hierarchy**

Maps constructed via the basic cartographic guidelines are not necessarily visually appealing maps. Although well-constructed, they might still look sterile. The design aspect of creating appealing maps also has to be included in the visualization process. 'Appealing' does not only mean having nice colours.

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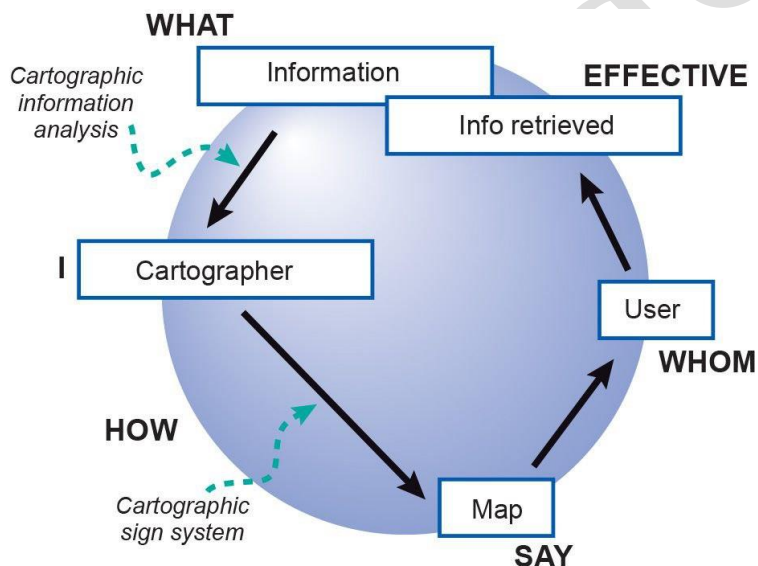
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7. What do you mean by "How do I Say What to Whom and is it effective?" in GIS? Explain. (NOV-19)(APR-23)

Ans:

The sentence “How do I say what to whom and is it effective?” guides the carto-graphic visualization process and summarizes the cartographic communication principle. Especially when dealing with maps in the realm of presentation cartograph, it is important to adhere to the cartographic design rules. This is to guarantee that the resulting maps are easily understood by their users. How does this communication process work? It starts with information to be mapped (the ‘What’ from the sentence). Before anything can be done, the cartographer should get a feel for the nature of the information, since this determines the graphical options. Cartographic information analysis provides this. Based on this knowledge, the cartographer can choose the correct symbols to represent the information in the map. S/he has a whole toolbox of visual variables available to match symbols with the nature of the data.



8. Explain visualization strategies in GIS.

(Nov-19)(Apr-23)

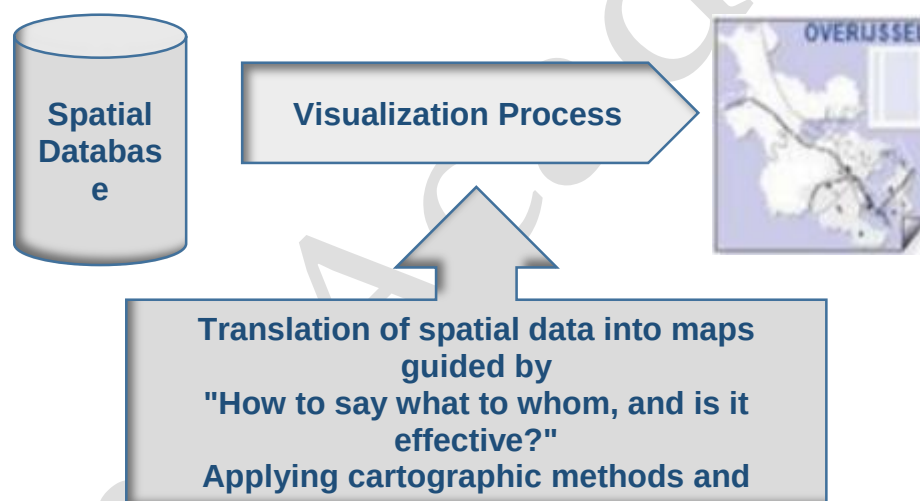
Ans:

The visualization process in GIS:

The characteristic of maps and their function in relation to the spatial data handling process was explained in the previous section. In this context the carto-graphic visualization process is the translation or conversion of spatial data from a database into graphics.

These are predominantly map-like products. During the visualization process, cartographic methods and techniques are applied. These can be considered to form a kind of grammar that allows for the optimal design and production for the use of maps, depending on the application.

The visualization process can vary greatly depending on where in the spatial data handling process it takes place and the purpose for which it is needed. Visualizations can be, and are, created during any phase of the spatial data handling process as indicated before. They can be simple or complex, while the production time can be short or long.



This thinking process is accelerated by the continued developments in hardware and software. Media like DVD-ROMs and the WWW allow dynamic presentation and also user interaction. These went along with changing scientific and societal needs for georeferenced data and maps. Users now expect immediate and real-time access to the data; data that have become abundant in many sectors of the geoinformation world. This abundance of data, seen as a paradise' by some sectors, is a major problem in other sectors. We lack the tools for user-friendly queries and retrieval when studying the massive amount of spatial data produced by sensors, which is now available via the WWW. A new branch of science is currently evolving to deal with this problem of abundance. In the geo-disciplines, it is called visual data mining.

These developments have given the term visualization an enhanced meaning. According to the dictionary, it means 'to make visible' or 'to represent in graphical form'. It can be argued that, in the case of spatial data, this has always been the business of cartographers. Progress in other disciplines has linked the word to more specific ways in which modern computer technology can facilitate the process of 'making visible' in real time. Specific software toolboxes have been developed, and their functionality is based on two key words: interaction and dynamics. A separate discipline, called scientific visualization, has developed around it, and has also had an important impact on digital cartography. It offers the user the possibility of instantaneously changing the appearance of a map. Interaction with the map will stimulate the user's thinking and will add a new function to the map. As well as communication, it will prompt thinking and decision-making.



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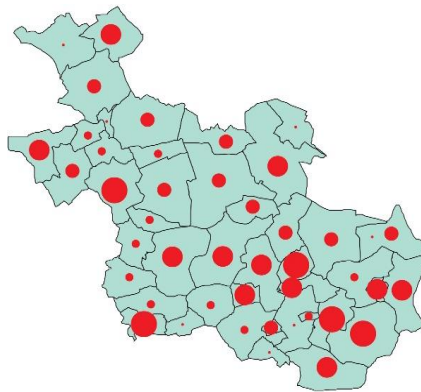
9. How to map quantitative data? Explain.

(Nov-19)

Ans:

Map quantitative data:

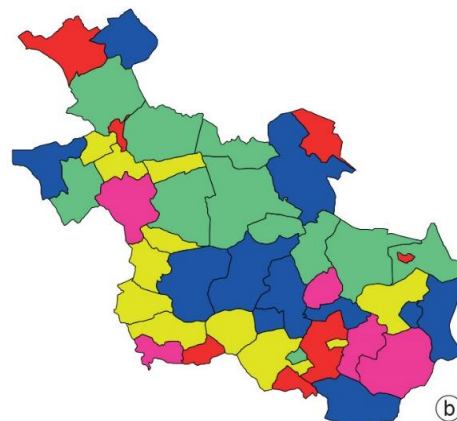
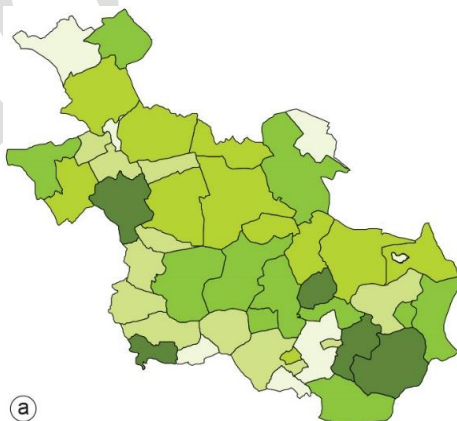
After executing a census, one would for instance like to create a map with the number of people living in each municipality, one deals with absolute quantitative data. The geographic units will logically be the municipalities. The final map should allow the user to determine the amount per municipality and offer an overview of the geographic distribution of the phenomenon. To reach this objective, the symbols used should have quantitative perception properties. Symbols varying in size fulfil this demand. Figure shows the final map for the province of Overijssel.



The reader might get a reasonable impression of the individual amounts but not of the actual geographic distribution of the population, as the size of the geographic units will influence the perception properties too much. Imagine a small and a large unit having the same number of inhabitants. The large unit would visually attract more attention, giving the impression there are more people than in the small unit. Another issue is that the population is not necessarily homogeneously distributed within the geographic units. Colour has also been misused in Figure(b). The applied four-colour scheme makes it impossible to infer whether red represents more populated areas than blue.

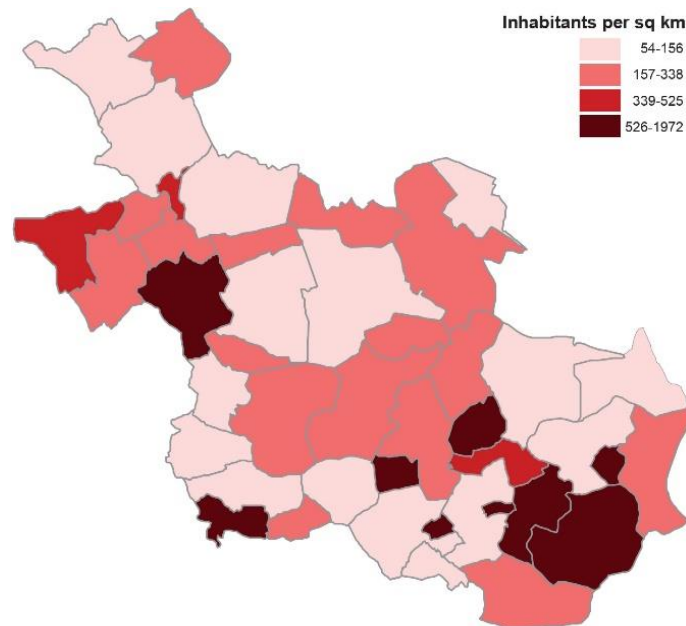
(a) wrong use of green tints for absolute population figures;

(b) incorrect use of colour



On the basis of absolute population numbers per municipality and their geographic size, we can also generate a map that shows population density per municipality. We then deal with relative quantitative data.

The aim of the map is to give an overview of the distribution of the population density from low(light tints) to high (dark tints). The map reader automatically and in glance associate the dark colours with high density and the light values with low density as shown in figure below



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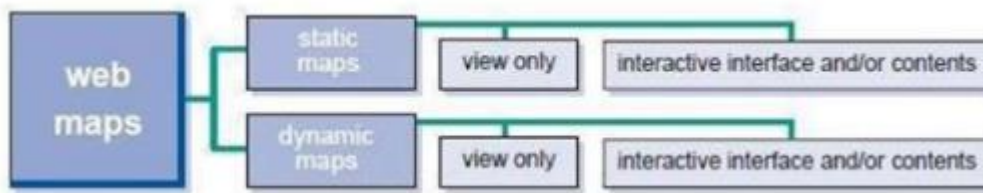
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10. Write a note on Map Disseminations.**(NOV-19)(APR-23)****Ans:****Map dissemination:****On-screen maps:**

Currently, most maps are presented on screen, for a quick view, for an internal presentation or for presentation on the WWW. Compared to maps on paper, on-screen maps must be smaller, and therefore their contents should be carefully selected. This might seem a disadvantage, but presenting maps on-screen offers very interesting alternatives. A mouse click could also open the link to a database, and reveal much more information than a paper map could ever offer.

**Multimedia maps:**

Maps and multimedia (photography, sound, video, and animation) can be integrated. Some of today's electronic atlases, such as the Encarta World Atlas are good examples of how multimedia elements can be integrated with the map. Pointing to a country on a world map starts the national anthem of the country or shows its flag. It can be used to explore a country's language; moving the mouse would start a short sentence in the region's dialects.

Static maps:

Organizations, such as map libraries or tourist information providers, often make their maps available in this way. This form of presentation can be very useful, for instance, to make historical maps more widely accessible. Static, view-only maps can also serve to give web surfers a preview of the products that are available from organizations, such as National Mapping Agencies.

Dynamic maps:

Dynamic maps are about change in one or more of the spatial data components. On the WWW, several options to play animations are available. The so-called animated-GIF can be seen as a view-only version of a dynamic map.

11. Define the following terms:**(APR-23)**

- i. **Symbology**
- ii. **Cartography**
- iii. **Map Legend**
- iv. **Pixel**
- v. **Voxel**

Ans:**i. Symbology**

In the context of Geographic Information Systems (GIS), symbology refers to the representation of geographic features on maps through the use of symbols. GIS is a technology that involves the capture, analysis, and visualization of spatial data, and symbology plays a crucial role in conveying information effectively on maps.

GIS symbology includes the design and application of symbols to represent different types of features such as points, lines, and polygons on a map. This representation helps users understand the characteristics and attributes of geographic features.

ii. Cartography

Cartography in GIS (Geographic Information Systems) refers to the process of creating, designing, and producing maps using spatial data. Cartography is an essential component of GIS as it involves the transformation of raw geographic data into visual representations that can be easily understood and interpreted. The goal of cartography in GIS is to communicate spatial information effectively through maps.

iii. Map Legend

In GIS (Geographic Information Systems), a map legend is a key or guide that provides an explanation of the symbols, colors, and patterns used on a map. The purpose of a map legend is to help users interpret the visual elements on the map and understand the meaning of different features or data represented. It serves as a reference tool, clarifying the significance of map symbols and aiding in the effective communication of spatial information.

iv. Pixel

In GIS (Geographic Information Systems), a pixel refers to the smallest unit of a digital image or raster dataset. The term "pixel" is a contraction of "picture element." In the context of GIS, raster datasets are used to represent continuous spatial phenomena, such as satellite imagery, aerial photographs, or elevation models.

v. Voxel

In GIS (Geographic Information Systems), a voxel is similar to a pixel, but it extends into the third dimension, representing a volumetric pixel. The term "voxel" is a combination of "volume" and "pixel." While pixels are two-dimensional units used in raster datasets, voxels represent the smallest unit of a three-dimensional raster dataset, often used to represent volumetric data.



12. How to map time series? Explain using suitable example.**(APR-23)****Ans:**

Mapping time means mapping change. This may be change in a feature's geometry, in its attributes or both. Examples of changing geometry are the evolving coastline of the Mumbai, the location of India's national boundaries, or the position of weather fronts. The changes of a land parcel's owner, landuse, or changes in road traffic intensity are examples of changing attributes. Urban growth is a combination of both. The urban boundaries expand and simultaneously the land use shifts from rural to urban. If maps are to represent events like these, they should be suggestive of such change.

1. Single static map

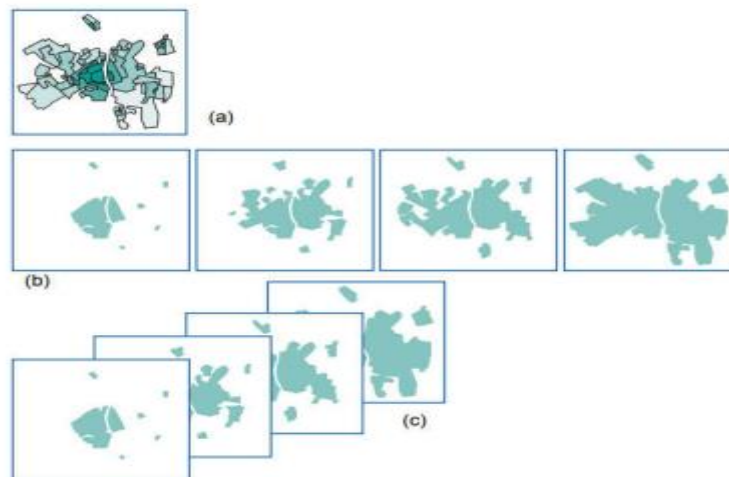
Specific graphic variables and symbols are used to indicate change or represent an event. Figure (a) applies the visual variable value to represent the availability of the water in Maharashtra;

2. Series of static maps

A single map in the series represents a 'snapshot' in time. Together, the maps depict a process of change. Change is perceived by the succession of individual maps depicting the situation in successive snapshots. It could be said that the temporal sequence is represented by a spatial sequence, which the user has to follow, to perceive the temporal variation. The number of images should be limited since it is difficult for the human eye to follow long series of maps.

3. Animated map

Change is perceived to happen in a single image by displaying several snapshots after each other just like a video cut with successive frames. The difference with the series of maps is that the variation can be deduced from real 'change' in the image itself, not from a spatial sequence.



13. Suppose one has two maps, one at scale 1:10,000 and another at scale 1:1,000,000. Which of the two maps can be called a large-scale map and which is a small scale map?

Ans:

The terms "large-scale" and "small-scale" in cartography can be a bit confusing because they are counterintuitive. In cartography:

Large-scale map: A large-scale map shows a small geographic area with great detail. The scale ratio is usually expressed as a fraction or ratio, such as 1:10,000. Therefore, in your example, the map at a scale of 1:10,000 is a large-scale map because it provides a more detailed representation of a smaller area.

Small-scale map: A small-scale map, on the other hand, depicts a larger geographic area but with less detail. The scale ratio is typically a smaller fraction, like 1:1,000,000. In this case, the map at a scale of 1:1,000,000 is a small-scale map because it covers a larger area with less detail compared to the 1:10,000 scale map.

So, to summarize:

1:10,000 is a large-scale map.

1:1,000,000 is a small-scale map.

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14. Describe the difference between a topographic map and a thematic map.**Ans:**

In GIS (Geographic Information System), both topographic maps and thematic maps serve different purposes and provide distinct types of information. Here's a brief description of the differences between the two:

Topographic Map:**Purpose:**

Topographic maps primarily focus on representing the physical features of the Earth's surface in detail. They provide a comprehensive view of natural and human-made features such as mountains, rivers, valleys, forests, roads, and settlements.

Content:

Topographic maps typically include contour lines to represent elevation changes, symbols for various physical features, and may also show land use and cover information. They aim to depict the three-dimensional landscape on a two-dimensional surface.

Scale:

Topographic maps often have larger scales (e.g., 1:24,000) to capture intricate details of a smaller geographic area.

Thematic Map:**Purpose:**

Thematic maps, on the other hand, are designed to represent specific themes or topics related to a geographic area. They focus on a particular subject or attribute rather than providing a comprehensive overview of the terrain.

Content:

Thematic maps can represent a wide range of topics, including population density, climate, land use, vegetation types, and more. These maps use various symbols, colors, or shades to convey information about the chosen theme.

Scale:

Thematic maps can have different scales depending on the scope of the theme being portrayed. They can cover small or large areas, and the scale is chosen to effectively communicate the thematic information.

In summary, while topographic maps emphasize the physical features of the landscape with a focus on detailed representation, thematic maps highlight specific themes or attributes related to a geographic area. Both types of maps play crucial roles in GIS and are used for different purposes in spatial analysis and decision-making.

15. Which four main types of thematic data can be distinguished on the basis of their measurement scales?

Ans:

Certainly! The four main types of thematic data distinguished on the basis of their measurement scales in GIS are nominal, ordinal, interval, and ratio data. Here's an explanation of each:

i. Nominal Data:

Explanation: Nominal data represent categories or labels with no inherent order or ranking. These categories are distinct and mutually exclusive, but they do not have a quantitative significance. Examples include land use types, categorical classifications, or feature types like rivers, lakes, and mountains.

ii. Ordinal Data:

Explanation: Ordinal data involve categories with a meaningful order or ranking, but the intervals between them are not consistent. The differences between values are not standardized or quantifiable. Examples include map legends with ordered classes such as low, medium, and high population density, where the order matters but the intervals may not be uniform.

iii. Interval Data:

Explanation: Interval data have ordered categories with consistent intervals between them. However, the absence of a true zero point means that ratios between values are not meaningful. Temperature measured in Celsius or Fahrenheit is an example of interval data because a temperature difference of 10 degrees has the same magnitude whether it's 10 to 20 or 20 to 30.

iv. Ratio Data:

Explanation: Ratio data have ordered categories with consistent intervals, and they possess a true zero point, making ratios between values meaningful. Examples include population counts, area measurements, or elevation. For instance, a population of 100 is twice the magnitude of a population of 50, indicating a meaningful ratio.

In summary, these four types of thematic data in GIS are differentiated based on the nature of their measurement scales, ranging from mere categories (nominal) to ordered categories with meaningful ratios (ratio). Understanding the scale of the data is crucial for selecting appropriate visualization methods and conducting meaningful spatial analyses.

16. Describe a number of ways in which a three-dimensional terrain can be represented on a flat map display.

Ans:

Representing three-dimensional terrain on a flat map in GIS involves various techniques to convey elevation, relief, and spatial relationships accurately. Here are several common methods for achieving this:

i. Contour Lines:

Description: Contour lines connect points of equal elevation, providing a visual representation of the terrain's shape. The spacing between contour lines indicates the steepness of the slope.

ii. Hillshading:

Description: Hillshading simulates the effect of light and shadow on the terrain, creating a 3D-like appearance. It enhances the perception of elevation by highlighting slopes facing the light source and shadowing those facing away.

iii. Digital Elevation Models (DEMs):

Description: DEMs are raster datasets that represent terrain elevation at regularly spaced intervals. They can be visualized using color gradients or shaded relief to depict elevation variations across the map.

iv. 3D Terrain Models:

Description: Advanced GIS software allows the creation of 3D terrain models that users can rotate, tilt, and zoom. These models provide an immersive experience and are helpful for visualizing terrain in a more interactive way.

v. Layer Stacking and Transparencies:

Description: Overlaying various map layers, such as elevation data, land cover, and other thematic information, while adjusting their transparency, can create a composite view that conveys both terrain features and additional context.

vi. Slope Maps:

Description: Slope maps use color gradients or patterns to represent the steepness of slopes across the terrain. Steeper slopes may be depicted with warmer colors or distinct patterns, aiding in the interpretation of elevation changes.

vii. 3D Symbols and Extrusions:

Description: Representing features like buildings, trees, or geological formations with 3D symbols or extrusions can enhance the visualization of terrain by adding vertical elements, helping to convey the three-dimensional nature of the landscape.

viii. Terrain Tinting:

Description: Applying color variations to the terrain based on elevation can provide a quick visual reference. For example, lower elevations may be represented in shades of green, while higher elevations are depicted in shades of brown or gray.

ix. Isometric or Oblique Views:

Description: Displaying the terrain from an oblique or isometric perspective instead of a traditional top-down view can offer a more realistic portrayal of the landscape.

x. Virtual Reality (VR) or Augmented Reality (AR):

Description: Integrating GIS with virtual reality or augmented reality technologies allows users to experience terrain in a more immersive and realistic way, providing a sense of depth and scale.

Choosing the appropriate method depends on the specific goals of the map and the audience's needs for understanding the terrain features.



17. Describe different techniques of cartographic output from the user's perspective.

Ans:

Certainly! From the user's perspective in GIS (Geographic Information System), various techniques of cartographic output are essential for effectively communicating spatial information. Here are some key techniques:

i. Interactive Web Maps:

Description: Users appreciate web-based maps that allow them to interactively explore spatial data, toggle layers, and access information dynamically through a browser, enhancing engagement and customization.

ii. Mobile Mapping Applications:

Description: GIS-enabled mobile apps provide users with on-the-go access to dynamic maps, offering features like GPS navigation, real-time location tracking, and data visualization on smartphones or tablets.

iii. Customized GIS Dashboards:

Description: Users benefit from customized GIS dashboards that integrate maps with other visualizations, enabling them to monitor real-time data, analyze trends, and make informed decisions through a user-friendly interface.

iv. Printed Maps and Atlases:

Description: Traditional printed maps and atlases remain valuable, providing users with tangible, portable references for navigation, fieldwork, or general spatial awareness.

v. Digital Story Maps:

Description: Story maps combine maps with narrative elements, images, and multimedia to guide users through a spatial story, enhancing communication and understanding of geographic information.

vi. 3D Visualizations:

Description: Users benefit from 3D visualizations that allow them to explore terrain, urban landscapes, or infrastructure models in three dimensions, providing a more immersive and realistic spatial experience.

vii. Augmented Reality (AR) and Virtual Reality (VR) Applications:

Description: AR and VR technologies offer users immersive experiences by overlaying digital map information onto the real world or providing virtual environments for in-depth spatial exploration.

viii. Geospatial Data Downloads:

Description: Users appreciate the ability to download geospatial datasets in various formats (e.g., shapefiles, GeoJSON), enabling them to use the data in their preferred GIS software for further analysis.

ix. Custom GIS Applications:

Description: Organizations or individuals may develop custom GIS applications tailored to specific user needs, offering specialized map interfaces or tools for data visualization and analysis.

x. PDF Maps and Electronic Documents:

Description: Users can access and share maps in electronic formats like PDFs, preserving interactive features, such as hyperlinks and layers, for easy viewing and printing.

xi. Print-on-Demand Services:

Description: Users can utilize print-on-demand services to create customized maps based on their specifications, allowing them to choose map extent, scale, and thematic content.

These techniques cater to diverse user preferences and requirements, providing flexibility in accessing, analyzing, and sharing spatial information in GIS.

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18. Explain the difference between static maps and dynamic maps.**Ans:**

The difference between static maps and dynamic maps in GIS lies in their level of interactivity and the ability to change or update in real-time. Here's an explanation of each:

~ Static Maps:**~ Definition:**

Static maps are fixed and unchanging representations of geographic information at a specific point in time. They are typically printed or saved as image files, such as JPEG or PNG.

~ Characteristics:

The content of static maps is predetermined and does not change unless a new version is created.

Users cannot interact directly with the map, zoom in/out, or toggle layers to reveal additional information.

Commonly used for traditional cartographic outputs, presentations, or hard-copy distribution.

Examples include paper maps, map posters, or digital images saved as snapshots of a particular map view.

~ Dynamic Maps:**~ Definition:**

Dynamic maps are interactive and can change in real-time or in response to user input. They are typically viewed through GIS software, web applications, or mobile mapping platforms.

~ Characteristics:

Dynamic maps allow users to interact with the data, enabling features like zooming, panning, and toggling layers on and off.

Information on dynamic maps can be updated in real-time or through scheduled updates, reflecting changes in the underlying data.

Commonly used for web mapping applications, GIS software interfaces, and other platforms where real-time or interactive capabilities are essential.

Examples include web maps that users can manipulate, GIS software interfaces with live data feeds, or applications displaying real-time traffic information.

In summary, the key distinction is that static maps are fixed representations that do not change, while dynamic maps offer interactivity and the ability to update in real-time, providing users with more flexibility and responsiveness in exploring and analyzing geographic information.

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